

Session Objectives

- Describe basic mechanisms of arrhythmia formation
- Explain the formation of reentry, triggered activity, and abnormal automaticity
- Identify common factors affecting arrhythmia formation
- Interpret common clinical arrhythmias and their underlying mechanisms
- Identify common cardiac channelopathies and analyze their mechanisms for arrhythmia formation

Definition of Arrhythmias

=not normal sinus rhythm

Normal heart beat:

1. Origin: Sinoatrial-SA node or the sinus node (SN)
2. Rate: 60-100bpm
3. Rhythm: regular
4. Conduction sequence
SN-Atria-AVN-His bundle (supraventricular)-
Bundle branches-Purkinje system-Ventricular myocytes

Any deviation from normal heart beat is arrhythmia

Note that some arrhythmias could be physiological

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Gaztanaga et al, Rev Esp Cardiol. 2012;65(2):174–185

Arrhythmia classifications (2)

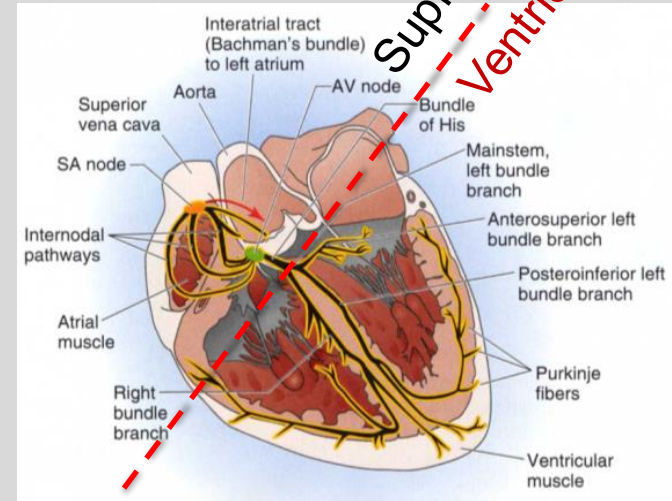
➤ For tachy-arrhythmias

1. Supraventricular arrhythmias

- originating from a site at His bundle or above
- with normal activation of ventricles (unless there is bundle branch block)
- typically with normal QRS (narrow)

2. Ventricular arrhythmias

- below His bundle
- abnormal ventricular activation
- widened QRS



Arrhythmia mechanisms

1. Abnormal Impulse Initiation

1). Abnormal automaticity

- a. Altered normal automaticity
- b. Abnormal automaticity

2). Triggered activity

- a. Early afterdepolarization
(EAD)
- b. Delayed afterdepolarization
(DAD)

2. Abnormal Impulse Conduction

1). Conduction block

common sites: SN-atrial,
AV node, His bundle &
bundle branches

2). Reentry

- a. Anatomic reentry
- b. Functional reentry

Altered Normal Automaticity

Increased (too fast,
>100bpm):

Sinus tachycardia-usually secondary, could be physiological (e.g., exercise), also seen in disease states (fever, thyrotoxicosis, etc.)

Inappropriate sinus tachycardia

Decreased (too slow,
<60bpm):

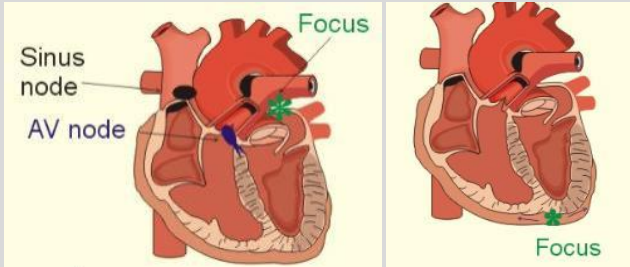
Sinus bradycardia-could be physiological (at rest), or in diseases (such as Sick Sinus Syndrome, SSS)

Abnormal Automaticity

Non-pacemaker myocyte (atrial & ventricular)

In disease state, could
gain pacemaker activity

Atrial arrhythmias
Ventricular arrhythmias



➤ Common factors promoting
abnormal automaticity:
hypoxia, ischemia, inflammation, etc.

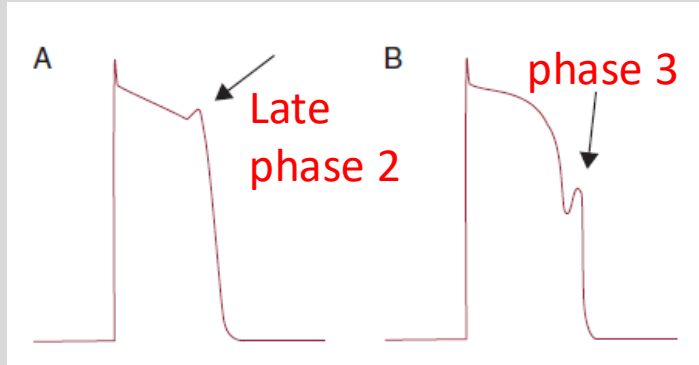
cause cell injury → membrane potential
increases (more positive) → inactivates
fast Na⁺ channels

could become slow response cells
(excites by I_{Ca-L})

Triggered Activity

(triggered by the previous AP)

1. Early afterdepolarization (EAD)

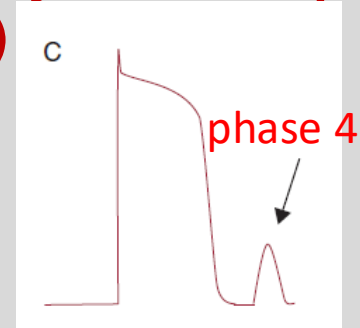


- Linked to prolonged AP duration: Long QT syndrome, hypokalemia, etc.
- due to re-opening of L-type Ca^{2+} Channel
- Risk $\uparrow$ at slower heart rate (bradycardia prolongs AP duration)

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2. Delayed afterdepolarization (DAD)



- Linked to Ca^{2+} overload: Digitalis toxicity, reperfusion-induced arrhythmia, CPVT (Catecholaminergic Polymorphic Ventricular Tachycardia) etc.
- Due to $\uparrow$ Na/Ca exchanger current
- Risk increases at faster heart rate

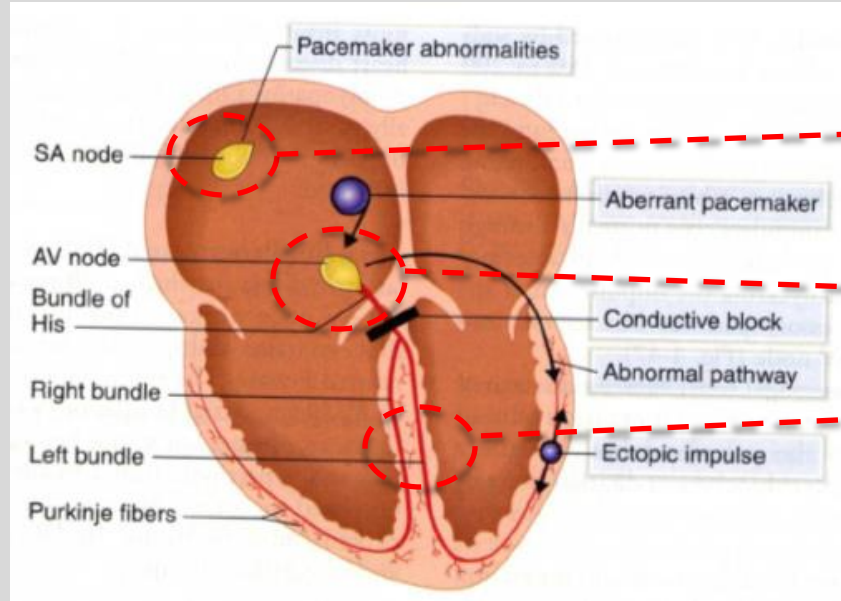
Abnormal Impulse Conduction (1)

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Conduction block (causes bradyarrhythmias)

Common conduction blocks:



Sino-atrial exit block
(sinus bradycardia or arrest)

→ AV (atrioventricular) block

→ Left or Right bundle branch
block (BBB)

Common clinical arrhythmias & mechanisms

Bradyarrhythmias

1. Failure of impulse generation
(**decreased automaticity**)
 - sinus bradycardia or arrest
(**Sick Sinus Syndrome, SSS**)
2. Failure of impulse propagation
(**conduction block**)
 - Sino-atrial block
 - AV block

Tachy-arrhythmias

(1 or more of these 3 basic mechanisms: abnormal automaticity, TA, or reentry)

1. Supraventricular tachyarrhythmias

originates at His bundle or above, normal/narrow-QRS, (with the exception of preexisting abnormal ventricular conduction)

2. Ventricular arrhythmias: (below His bundle, wide-QRS)

Supraventricular Tachyarrhythmias

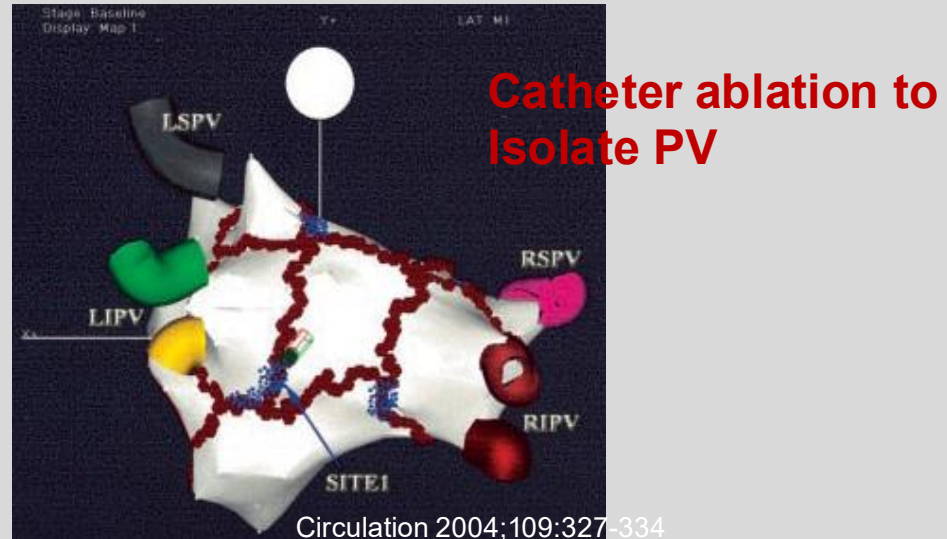
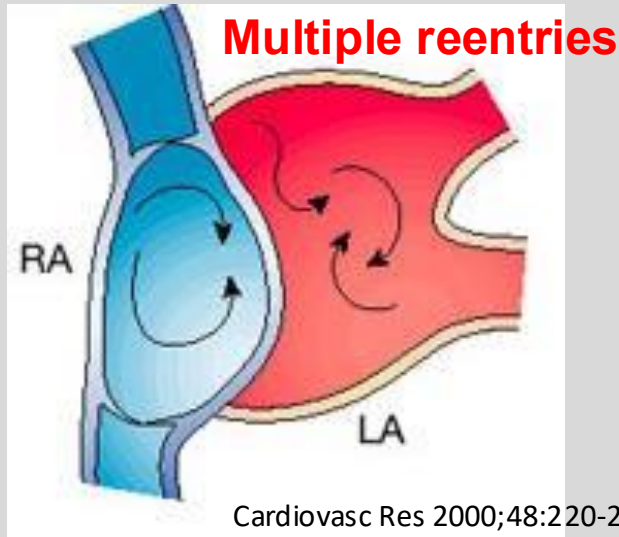
Clinical arrhythmia	Mechanisms
Sinus tachycardia	Enhanced normal automaticity
Atrial premature contraction	Abnormal automaticity, TA, or reentry
Atrial tachycardia	Abnormal automaticity, TA, or reentry
Atrial flutter	Reentry
Atrial Fibrillation	Multiple reentries (ectopic foci + substrate)
AVNRT (AV nodal reentrant tachycardia)	Reentry
AVRT (atrioventricular reentrant tachycardia)	Reentry
Atrioventricular junctional tachycardia	Enhanced normal automaticity, abnormal automaticity or TA

Atrial Fibrillation

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- The most common clinically significant arrhythmia
- **multiple reentries in both atria, requiring**
 - triggers or ectopic foci (abnormal automaticity, TA) and
 - substrate (remodeled atria, dilatation, fibrosis, to support multiple reentries).
- Most often triggers coming from the pulmonary veins (PV), that is why PV isolation is used to cure AF



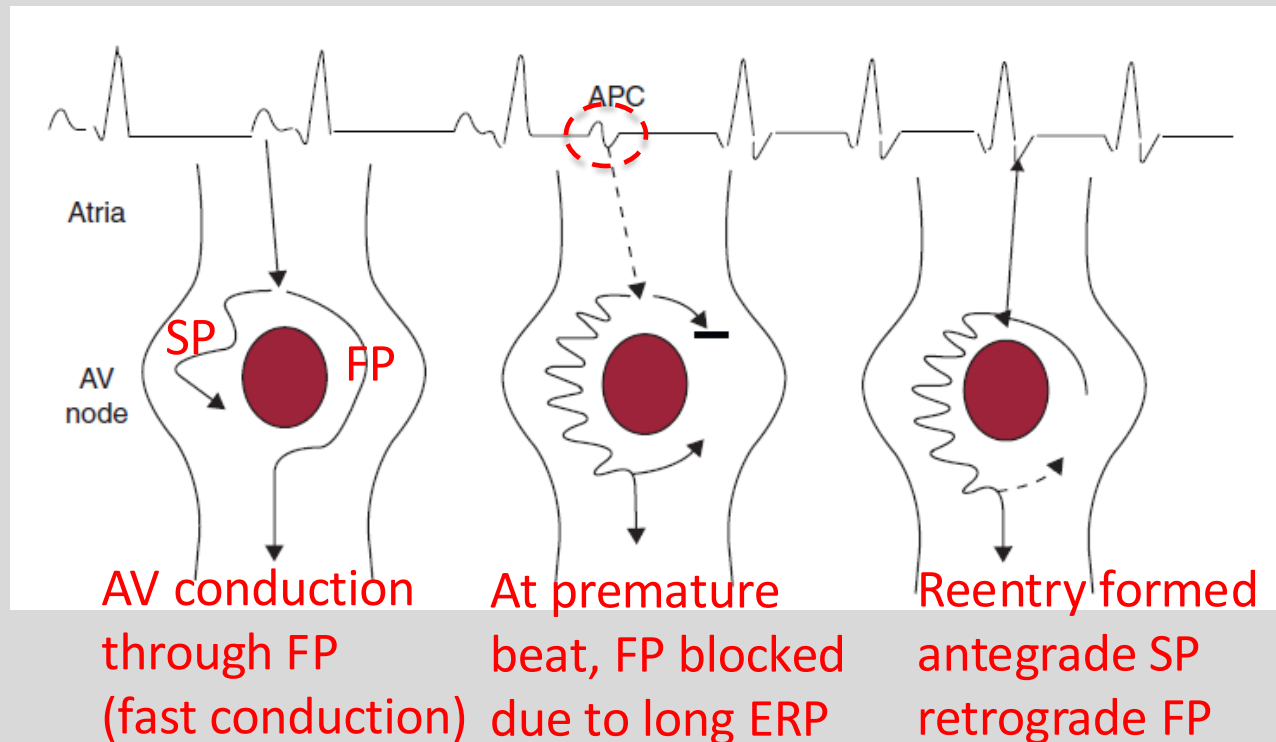
AVNRT

Atrioventricular Nodal Reentrant Tachycardia

reentry

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➤ AVN dual pathway electrophysiology

➤ SP ablation to cure AVNRT

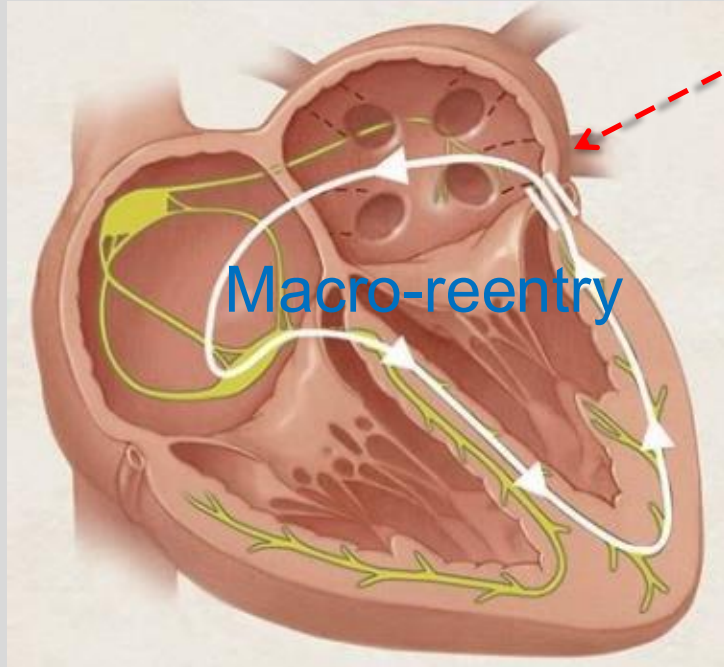
Micro-reentry

AVRT

Atrioventricular Reentrant Tachycardia
(Wolf-Parkinson-White (WPW) syndrome
Or Pre-excitation Syndrome)

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Accessory pathway (**Bundle of Kent**)
Seen on ECG (WPW)
concealed (no WPW on ECG)
Some may have multiple pathways
Cured by accessory pathway
ablation

If antegrade conduction through AVN-His-Purkinje system, then the tachycardia is with narrow-QRS.

If the reentry is in a reversed direction, then tachycardia is with widened-QRS.

Minimum requirement

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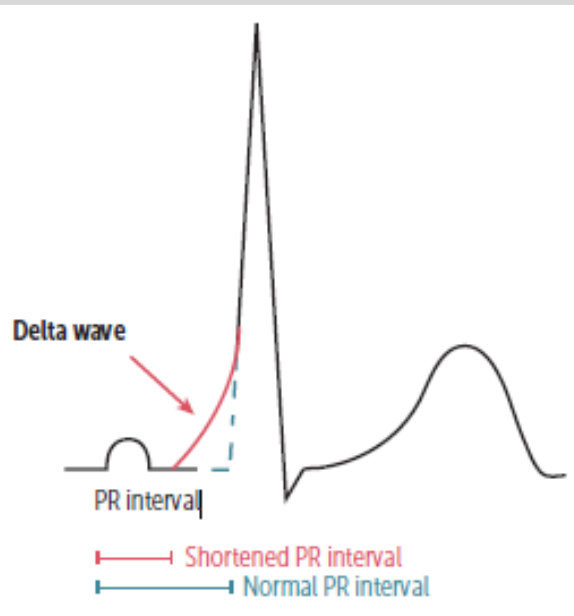
Wolff-Parkinson-White(WPW) syndrome:

Most common type of ventricular preexcitation syndrome.

Abnormal fast accessory conduction pathway from atria to ventricle (bundle of Kent) bypasses the rate-slowng AV node → ventricles begin to partially depolarize earlier → characteristic delta wave with widened QRS complex and shortened PR interval.

May result in reentry circuit

→ supraventricular tachycardia.



Minimum requirement

Congenital long QT syndrome

Inherited disorder of myocardial repolarization, typically due to ion channel defects; risk of sudden cardiac death (SCD) due to torsades de Pointes (**TDP-polymorphic VT**).

Includes:

Romano-Ward syndrome—autosomal dominant, pure cardiac phenotype (no deafness).

Jervell and Lange-Nielsen syndrome—autosomal recessive, sensorineural deafness.

Minimum requirement

Acquired LQT syndrome

Drug-induced long QT (**ABCDE**):

AntiArrhythmics (class IA, III)

AntiBiotics (e.g., macrolides)

Anti“C”ychotics (e.g., haloperidol)

AntiDepressants (e.g., TCAs)

AntiEmetics (e.g., ondansetron)

Polymorphic ventricular tachycardia (torsades de pointes, TDP), characterized by shifting sinusoidal waveforms on ECG; can progress to ventricular fibrillation.

Long QT interval predisposes to torsades de pointes.

Caused by **drugs**, $\downarrow K^+$, $\downarrow Mg^{2+}$, other abnormalities.

Treatment includes magnesium sulfate.

(4) CPVT

Catecholaminergic polymorphic VT: due to
leaky cardiac ryanodine receptor (RyR2)-**DAD**

genetic disorders of Ca^{++} release channels (RyR2) and their
controlling proteins (e.g. **calquestrin**)

Typically seen in structurally
normal heart

Sympathetic excitation like exercise or
acute emotion could trigger CPVT

**β -blocker therapy is used to prevent
CPVT**

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